Categories and Labels

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Categories and Labels before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a **label**led second line to correct your explanation.

Sound Becomes a Signal

Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Rules or Learned Patterns?

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

AI or Not?

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

Complete the supported Categories and Labels trace and record the first step you can explain without help.

Use W2: Replace a vague instruction for Categories and operational **label**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Categories and Labels explanation against the supplied facts before discussing it.

Use the worked case for Categories and operational **label**s. Proposed explanation: “A blank is the same as a failing value”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Transfer the method to a new record

A fictional equipment-request form accepts a whole-number borrowing duration of 1 through 7 days inclusive and one item code X or Y.

Label revise if any known field fails; otherwise **label** review if a field is blank or unreadable; otherwise **label** ready. Classify these forms and justify each: 7 days X; 8 days Y; unreadable duration X; 8 days blank code.

Explain whether ready alone grants permission to borrow.

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Categories and Labels target to a trace and a justified conclusion.

Your dataset **label**s 'reusable containers'. One team includes a disposable bottle reused once; another excludes it. Give an operational **label** rule and classify this boundary case under that rule.

Explain why a different rule could be defensible.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Categories and Labels case and show the deciding evidence.

Six cards show two circles, two triangles and two squares. Sort by shape.

> Why can red and blue circles share a **label**?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

> The rule is: fruit has seeds; other objects go in other. Sort an apple, a pencil and a sealed mystery box. Which item needs more **evidence**, and why?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 7 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Categories and Labels adds to Helpful Classroom Sorter and keep the evidence.

Define the Classroom Sorter's bin **label**s and write how to handle objects that reasonably fit more than one bin.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Your dataset **label**s 'reusable containers'. One team includes a disposable bottle reused once; another excludes it. Give an operational **label** rule and classify this boundary case under that rule.

Explain why a different rule could be defensible.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

W1 Apply the calendar guide

> Apply the calendar guide. Apply the supplied Categories and Labels facts and keep your reasoning.

The calendar policy accepts whole-number days 1 through 30 inclusive and a room code A or B. Check known failures first, then unknown fields, then readiness.

Label these cards and give **evidence**: day 30 room B; day 31 room A; day 14 room unreadable; day 31 room unreadable.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Replace a vague instruction

> Replace a vague instruction. Apply the supplied Categories and Labels facts and keep your reasoning.

A helper says **Label** a card ready if it looks nice. Explain why this can lead to inconsistent **label**s.

Replace it with an instruction that another person can apply using the calendar policy, including what to do about a blank field.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Count and investigate disagreement

> Count and investigate disagreement. Apply the supplied Categories and Labels facts and keep your reasoning.

For the same four cards, Mina records ready, revise, review, ready. Theo records ready, review, review, revise. How many positions agree? Identify the **disagreement**s.

Can you decide which **annotator** is correct from these **label** lists alone? Explain what else you need.

> **My reasoning and evidence**

WORKBOOK · 15 OF 21

W4 Transfer the method to a new record

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Transfer the method to a new record. Apply the supplied Categories and Labels facts and keep your reasoning.

A fictional equipment-request form accepts a whole-number borrowing duration of 1 through 7 days inclusive and one item code X or Y.

Label revise if any known field fails; otherwise **label** review if a field is blank or unreadable; otherwise **label** ready. Classify these forms and justify each: 7 days X; 8 days Y; unreadable duration X; 8 days blank code.

Explain whether ready alone grants permission to borrow.

> **My reasoning and evidence**

W5 Keep a policy change visible

> Keep a policy change visible. Apply the supplied Categories and Labels facts and keep your reasoning.

Version 1 accepts calendar days 1 through 30. Version 2 accepts days 1 through 31. A card says day 31 room A. State its **label** under each version.

Explain why mixing the two **label**s without recording the version could create a misleading **training** dataset.

> **My reasoning and evidence**

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

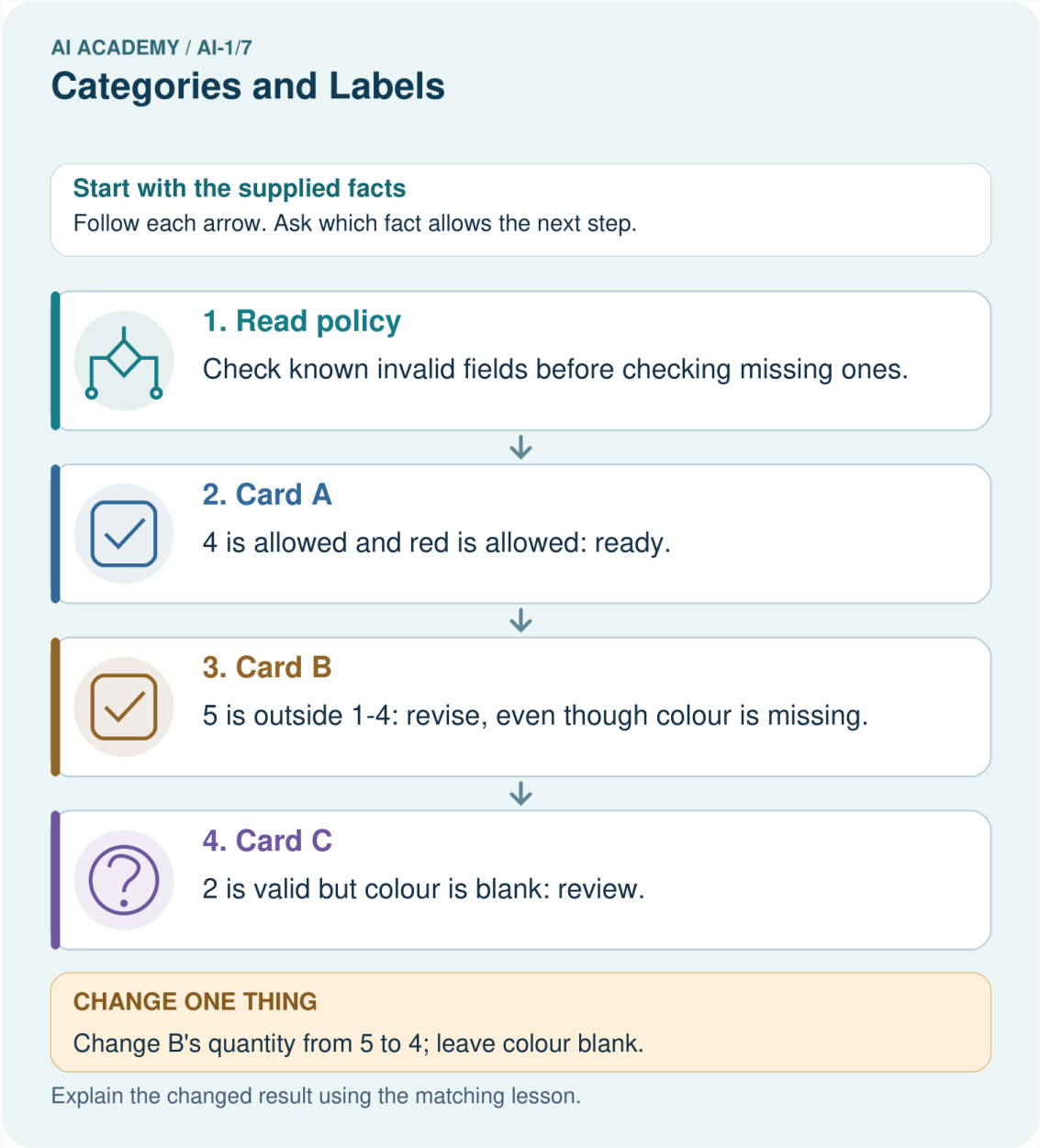
Pictorial case practice: Categories and Labels

> A fictional craft club labels cardboard-order cards ready, review or revise. Explain the labelled step before checking the changed condition.

A fictional craft club **label**s cardboard-order cards ready, review or revise.

Allowed quantity: whole number 1–4. Required colour: red or blue.

If any known field fails, revise; otherwise if a required field is blank, review; otherwise ready. Cards A=(4,red), B=(5,blank), C=(2,blank).



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why B originally was revise rather than review, then **label** its changed version.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Write a sorter **label** guide with examples of complete, invalid and unresolved **input**s before comparing **label**s.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Write a sorter **label** guide with examples of complete, invalid and unresolved **input**s before comparing **label**s.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional craft club **label**s cardboard-order cards ready, review or revise.

Practice 1: explain the deciding evidence

Which **label**led step matches this detail? 2 is valid but colour is blank: review.

1. Read policy

2. Card B

3. Card A

4. Card C

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Card C 2 is valid but colour is blank: review. Operational **label**s specify what a word means for this task.

Two **label**lers using the same priority should agree on these cards. Review describes missing **evidence**, not a hidden assumption that the card is wrong. Revise is supported by a known failure.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change B's quantity from 5 to 4; leave colour blank. Which conclusion is supported by the supplied facts?

1. B becomes review.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: B becomes review. The known failure is removed, but a required field is still missing.

Repair a missing building block

Review Sound Becomes a Signal (ai-1/6). Goal: Interpret an ordered sequence of sound-signal samples with a stated time interval. Short practice: Does the all-zero retained record prove the full signal was silent?

Use the picture.. Return to this week and explain what you changed.

Review Rules or Learned Patterns? (ai-1/4). Goal: Execute a stated rule including its boundary. Short practice: Show which stage used **label**s and which stage used only the new measurement..

Return to this week and explain what you changed.

Review AI or Not? (ai-1/1). Goal: Distinguish an observed result from an explanation of how it was produced. Short practice: Point to the learned stage. Does the changed display prove the weather forecast changed?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Categories and Labels picture from its labels, then record a relevant limit.

01. Categories and Labels: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

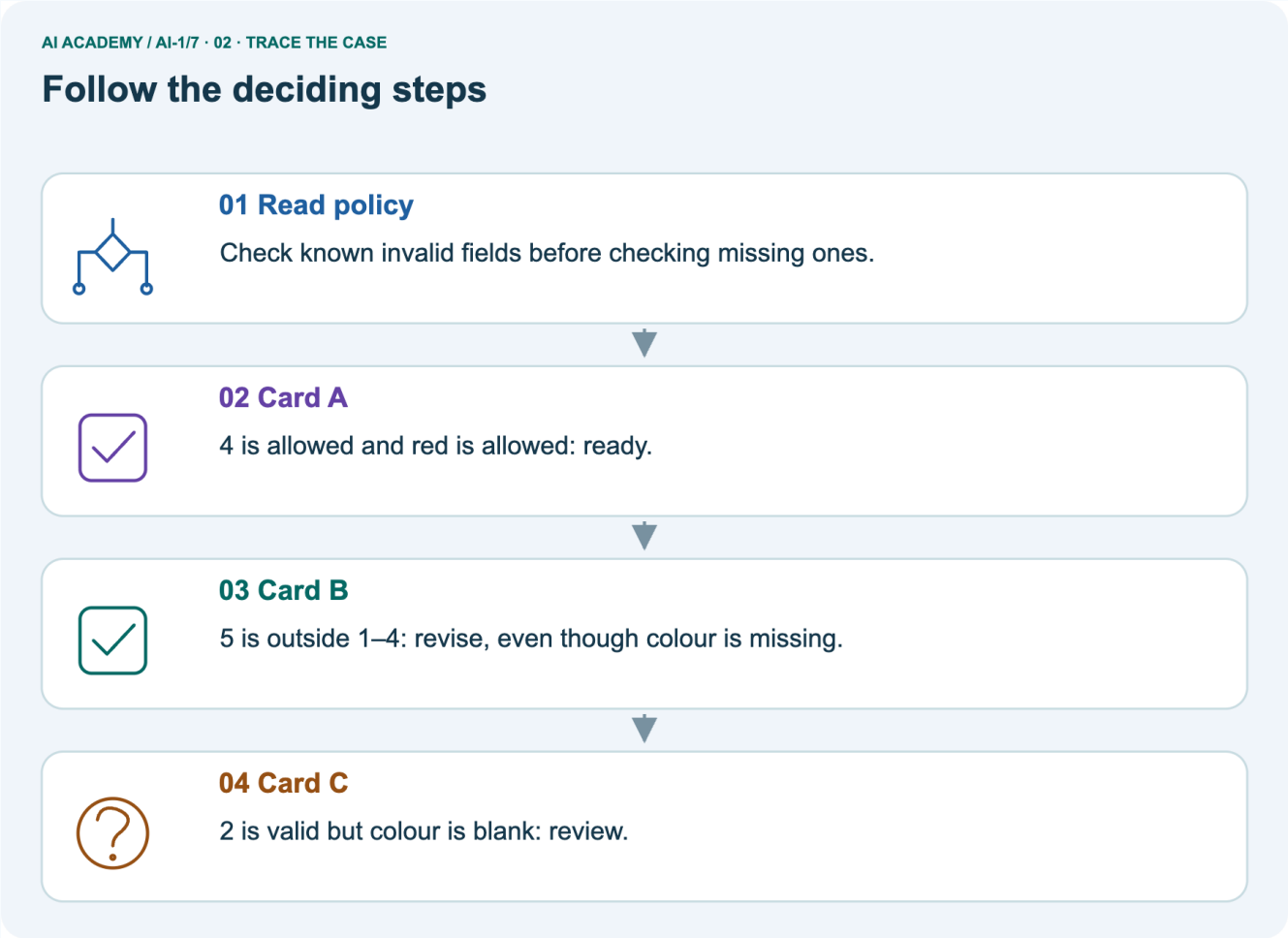
A fictional craft club **label**s cardboard-order cards ready, review or revise. Supplied facts: Allowed quantity: whole number 1–4. Required colour: red or blue.

If any known field fails, revise; otherwise if a required field is blank, review; otherwise ready. Cards A=(4,red), B=(5,blank), C=(2,blank).

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read policy: Check known invalid fields before checking missing ones. → Card A: 4 is allowed and red is allowed: ready. → Card B: 5 is outside 1–4: revise, even though colour is missing. → Card C: 2 is valid but colour is blank: review.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Three ordered checks for a calendar card



Follow the labelled picture with the matching explanation. Open the vector diagram

Three ordered checks for a calendar card. First check known values | Any known field fails its criterion | Revise even if another field is unknown.

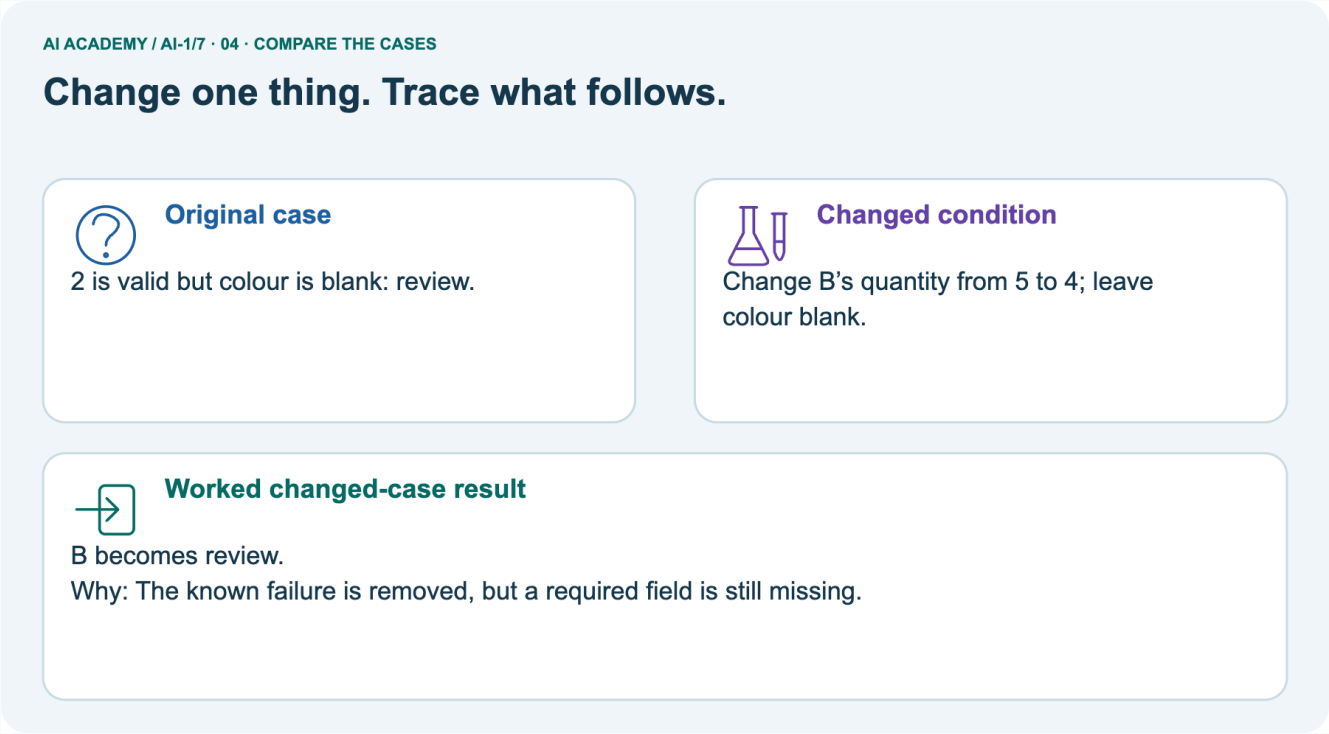
Then check missing information | No known failure but a field is blank or unreadable | Review because **evidence** is incomplete.

Otherwise all checks have passed | Day is 1 through 30 and room is A or B | Ready under this record policy

Read the **label**s and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

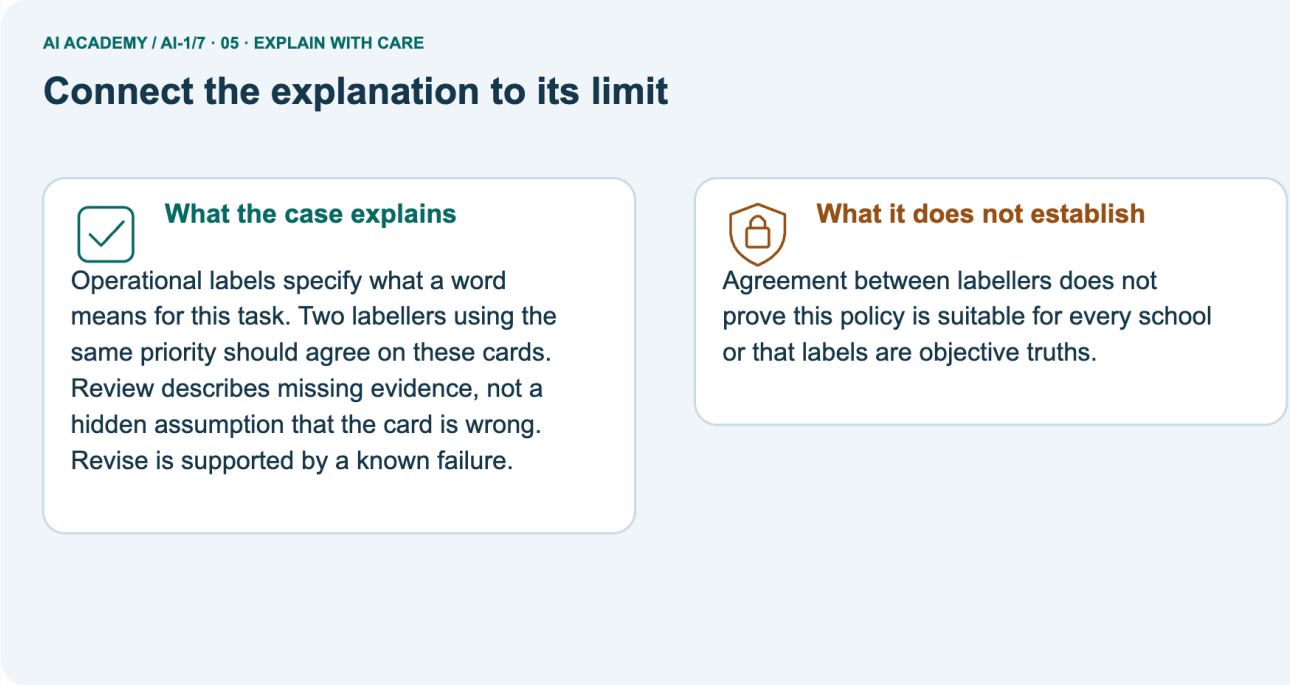
Original case: 2 is valid but colour is blank: review. Changed condition: Change B's quantity from 5 to 4; leave colour blank. Expected result: B becomes review.

Reason: The known failure is removed, but a required field is still missing.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

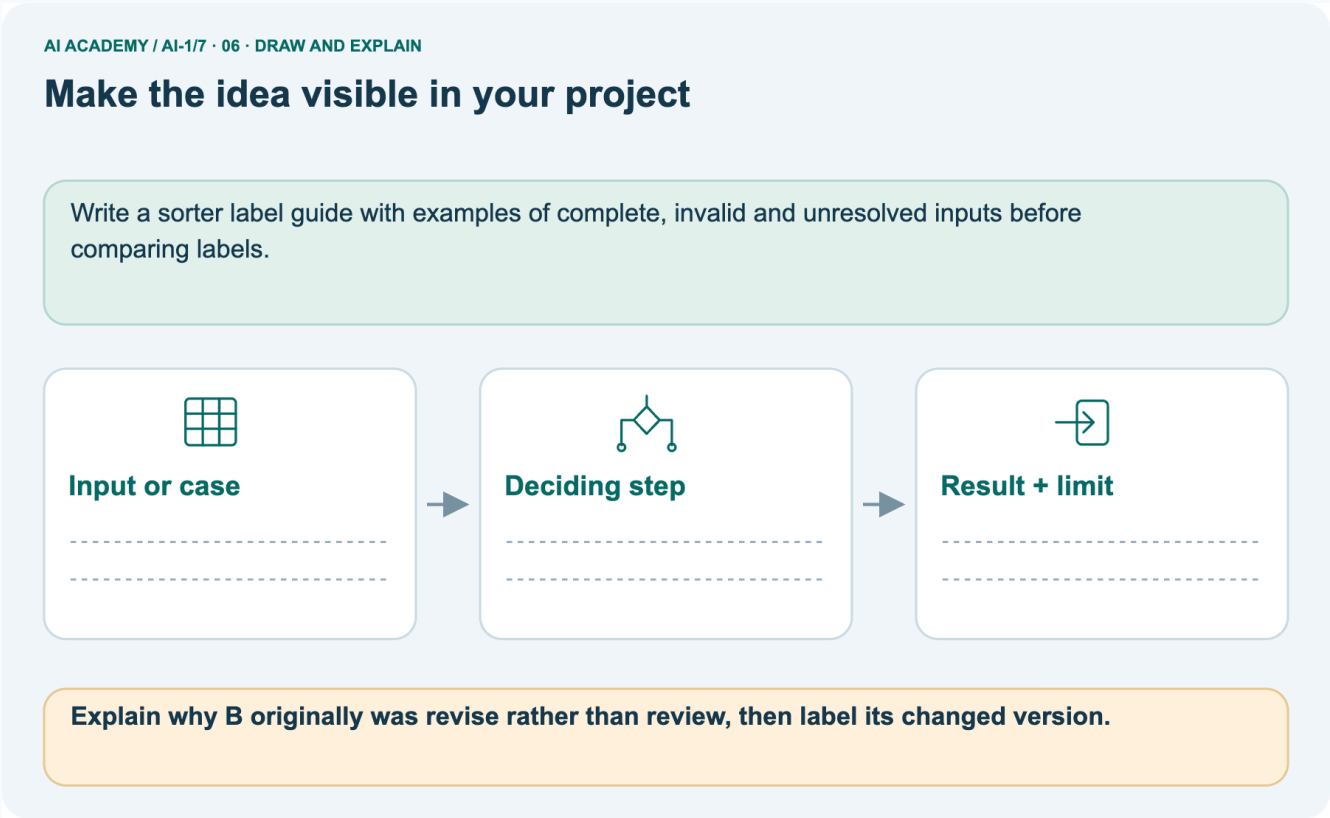
Operational **label**s specify what a word means for this task. Two **label**lers using the same priority should agree on these cards. Review describes missing **evidence**, not a hidden assumption that the card is wrong.

Revise is supported by a known failure. Limit: Agreement between **label**lers does not prove this policy is suitable for every school or that **label**s are objective truths.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Write a sorter **label** guide with examples of complete, invalid and unresolved **input**s before comparing **label**s. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B originally was revise rather than review, then **label** its changed version.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.